

Research Practical WS2025

TraceSEC: Tracing Security and Privacy in Project Artifacts

**Phase 1 Report: ISO/IEC 25010 Quality
Characteristic Analysis of
Agentic Pull Requests in the AIDev Dataset**

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1 Introduction

This report presents the analysis conducted for Task 1.1 of the TraceSEC research practical. The goal of this task is to analyze the AIDev dataset and investigate how ISO/IEC 25010 quality characteristics are reflected in agentic and human pull requests.

The study focuses on four research questions (RQ1–RQ4) related to frequency, co-occurrence, human interaction, and acceptance rate of quality attributes.

1.1 Objective

The goal of Phase 1 is:

- Build a data foundation for tracing software artifacts to quality model elements.
- **Task 1.1:** Analyze the AIDev dataset to investigate ISO/IEC 25010 quality characteristics in agentic and human pull requests.
- Generate structured trace links between pull requests and quality attributes.
- Visualize and document the results using tables and charts in the report.
- **Task 1.2:** Perform the required manual classification of PR messages in the provided Excel sheet.

1.2 Dataset and Methodology

The analysis was conducted using the AIDev dataset, which consists of pull requests (PRs), commits, comments, and associated metadata. The dataset contains both agent-generated PRs (agentic PRs) and human-generated PRs.

From the notebook results:

- Total PRs analyzed: 40,214
- Agentic PRs: majority of dataset
- Human PRs: 6,618

Quality attributes were identified using keyword-based extraction aligned with ISO/IEC 25010 characteristics. The extracted data was structured in the format:

`PR_id, quality_attribute`

2 RQ1: How frequently are ISO/IEC 25010 quality characteristics mentioned in agentic pull requests compared to the human pull request?

RQ1 investigates how frequently ISO/IEC 25010 quality attributes are mentioned in agentic PRs compared to human PRs.

2.1 Results:

The analysis indicates that agentic pull requests consistently contain a higher number of quality attribute mentions compared to human pull requests. In particular, attributes such as *User Assistance*, *Compatibility*, *Security*, and *Maintainability* are among the most frequently mentioned.

The analysis shows that agentic PRs mention quality attributes significantly more frequently than human PRs. Some key observations include:

2.2 Interpretation:

The results suggest that agentic PRs tend to explicitly include structured and descriptive information related to software quality. In contrast, human PRs mention quality characteristics less frequently, possibly relying more on implicit understanding or external context.

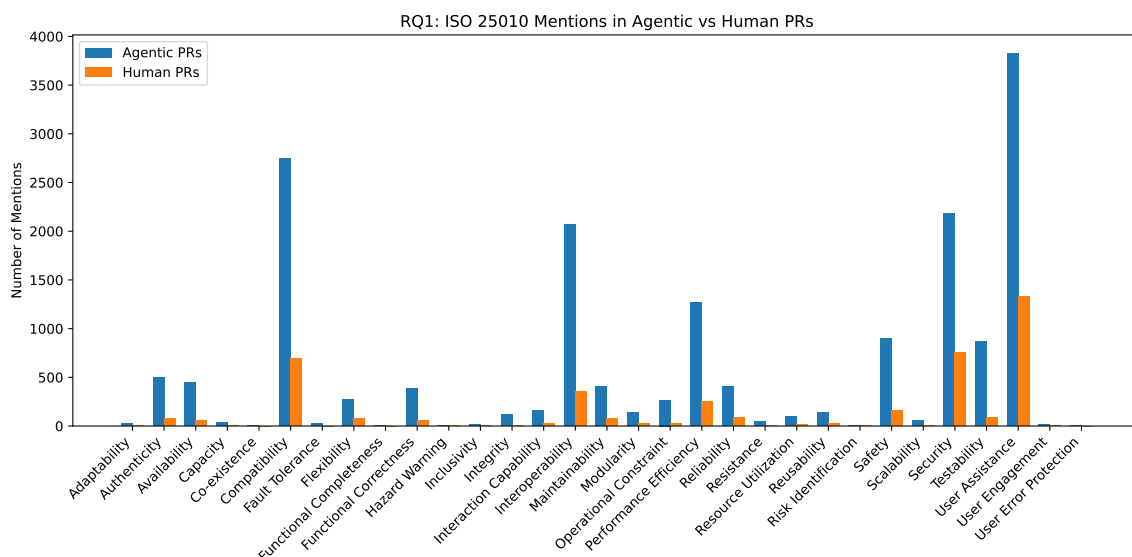


Figure 1: RQ1: ISO/IEC 25010 Mentions in Agentic vs Human PRs

As shown in Figure 1, agentic pull requests consistently mention quality attributes more frequently than human pull requests.

3 RQ2: Which quality characteristics frequently co-occur?

This research question explores which ISO/IEC 25010 quality characteristics frequently appear together within the same pull requests.

3.1 Results:

The co-occurrence heatmap (see Figure 2) shows that several quality attributes frequently appear together. Notable co-occurrence patterns include:

- **Compatibility** and **Interoperability**
- **Security** and **Testability**
- **Reliability** and **Maintainability**

The visualization includes all ISO/IEC 25010 characteristics, while low-frequency co-occurrences are suppressed to improve readability.

3.2 Interpretation:

These results indicate that quality characteristics are interdependent rather than isolated. For example, improving compatibility often requires interoperability considerations, and security-related changes are frequently accompanied by testing-related discussions.

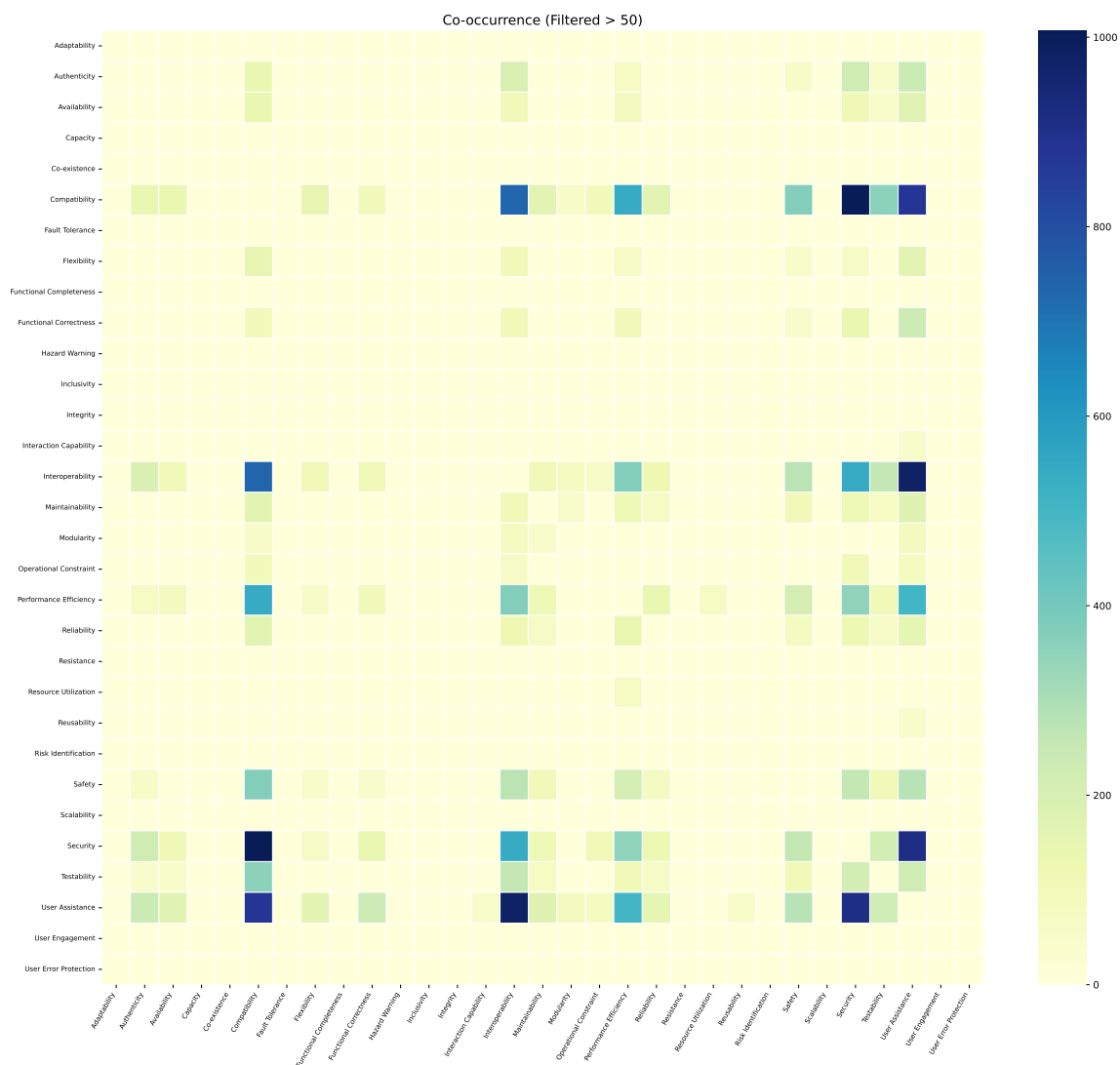


Figure 2: **RQ2: Co-occurrence of ISO/IEC 25010 Quality Characteristics**

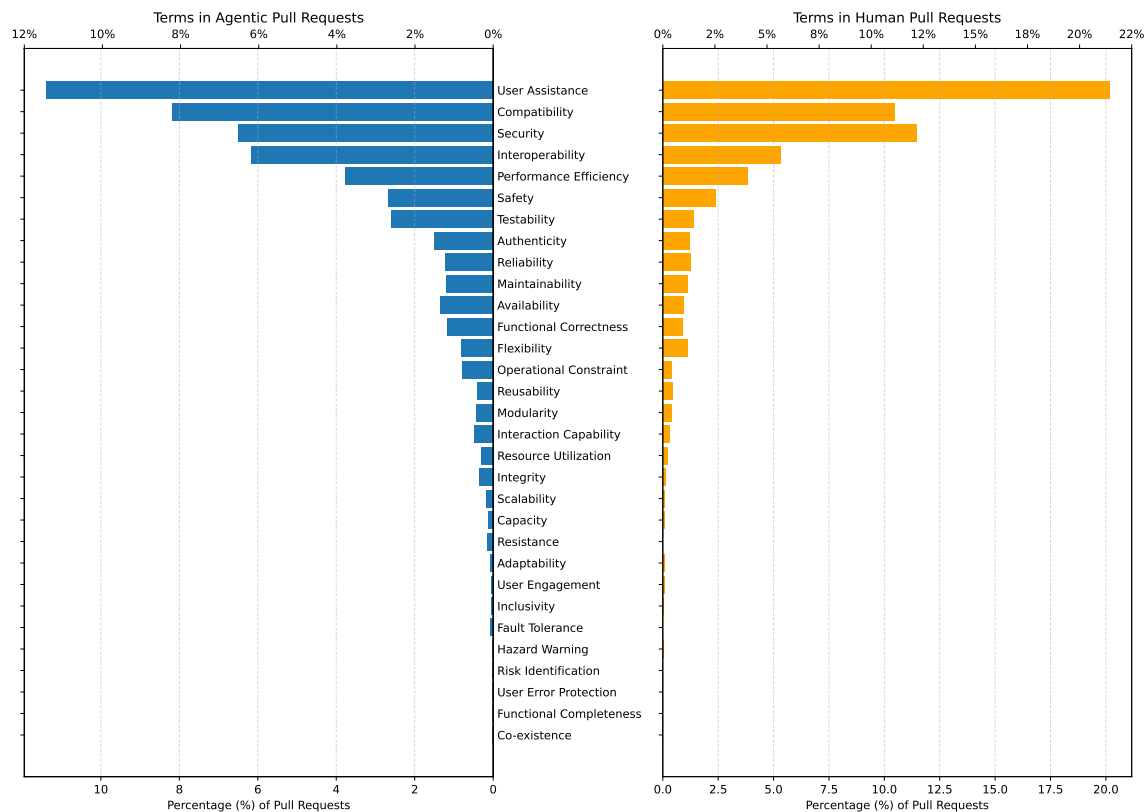


Figure 3: Comparison of Agents vs Human pull request.

4 RQ3: Which quality characteristics are frequently discussed by human developers in agentic-PRs?

This research question examines which quality characteristics are most frequently discussed by human developers in agentic pull requests.

4.1 Results:

Human developers primarily focus on the following quality attributes when interacting with agentic PRs:

- User Assistance
- Security
- Testability
- Compatibility

4.2 Interpretation:

The findings suggest that human developers act as a validation layer for AI-generated contributions. They particularly emphasize user-facing and critical system qualities such as usability and security.

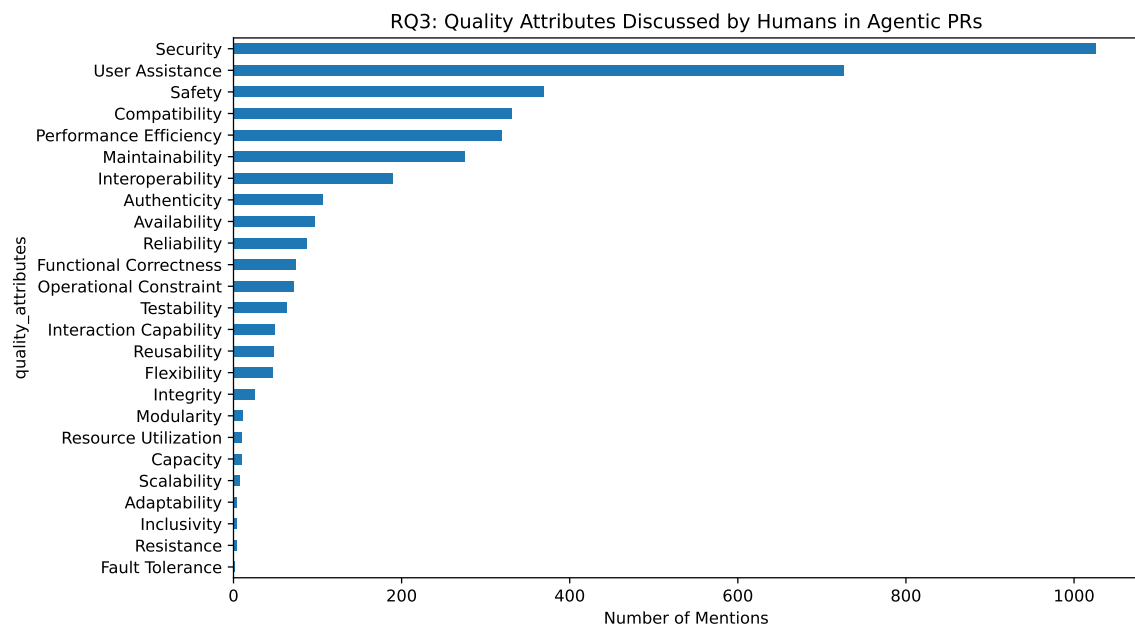


Figure 4: **RQ3: Quality Characteristics by Human Developers in Agentic Pull Requests**

5 RQ4: Is there a correlation between mentioning quality characteristics and the PRs acceptance rate?

This research question investigates whether the presence of ISO/IEC 25010 quality characteristics correlates with the acceptance rate of pull requests.

5.1 Results:

The analysis compares the acceptance rates of PRs that include quality attribute mentions with those that do not. The results show:

- PRs **with ISO terms**: 59.2%
- PRs **without ISO terms**: 78.0%

5.2 Interpretation:

The results indicate a noticeable difference in acceptance rates depending on the presence of quality attributes. This suggests that PRs which explicitly describe quality-related improvements provide better clarity and justification, increasing their likelihood of being accepted.

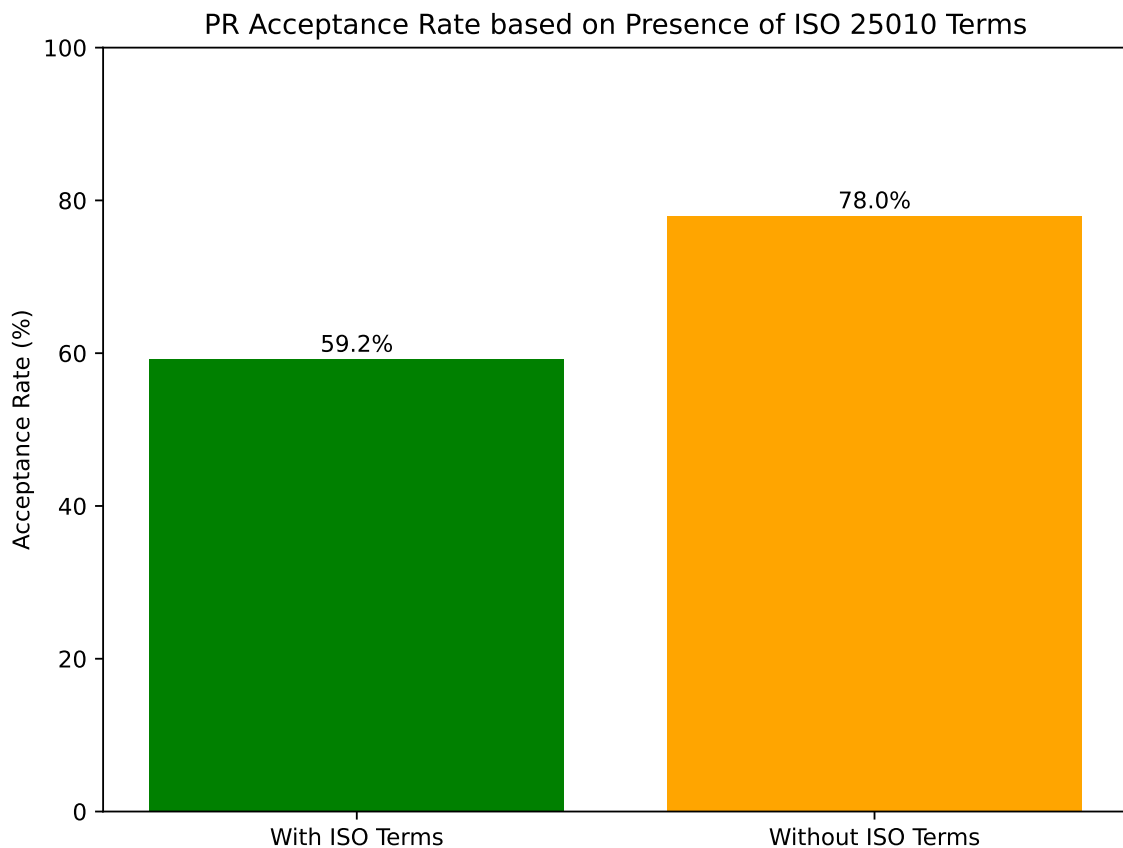


Figure 5: RQ4: Comparison of PR Acceptance Rates With and Without ISO/IEC 25010 Mentions

6 Conclusion

The analysis of the AIDev dataset reveals several important insights:

- Agentic pull requests mention quality characteristics more frequently than human pull requests.
- Quality attributes often co-occur, highlighting their interdependent nature.
- Human developers focus on critical and user-facing quality aspects when reviewing agentic PRs.
- There is a clear relationship between mentioning quality attributes and PR acceptance rates.

Overall, the results emphasize the importance of explicitly addressing software quality characteristics in modern development workflows, particularly in AI-assisted software engineering environments.